

Structured Loss Amplification for U-Net-based Human-scale Dark-field CT Streak Reduction

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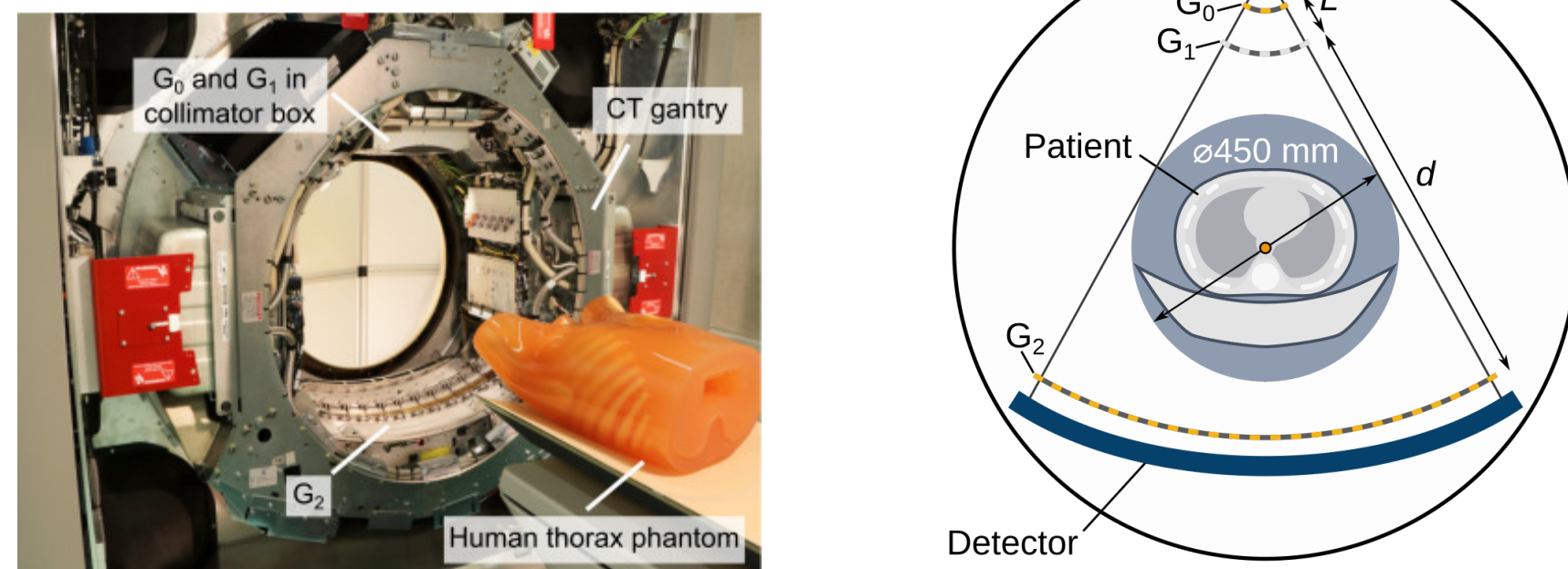
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INTRODUCTION

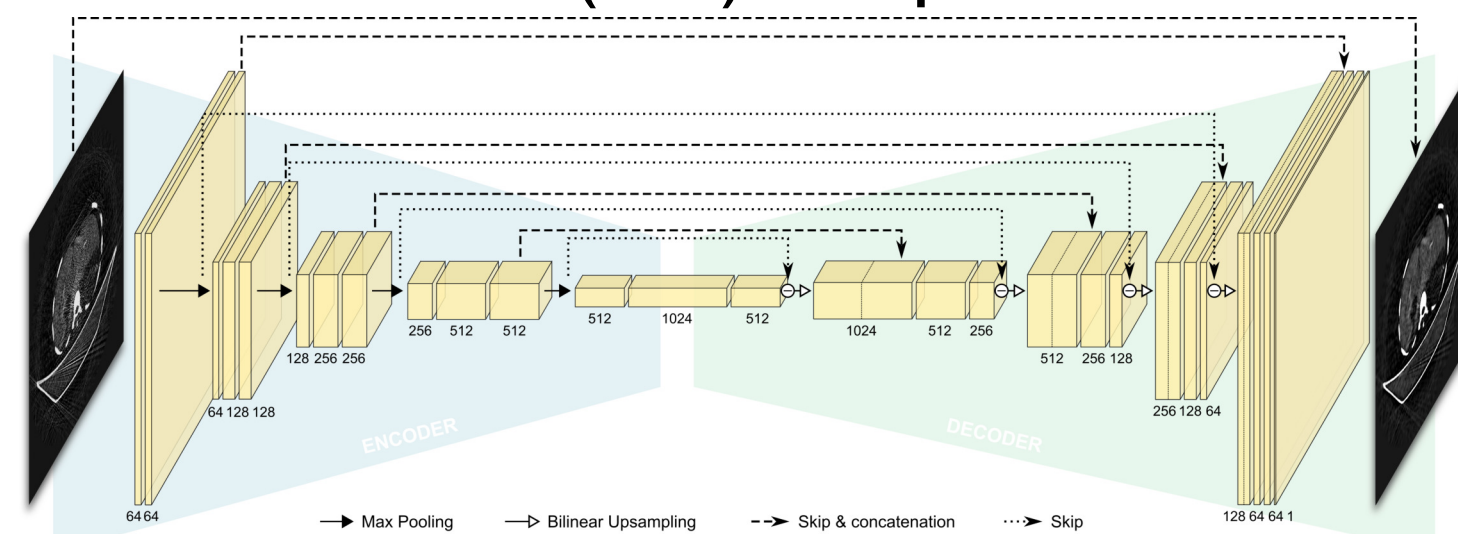
- **Dark-field Computed Tomography (DFCT)** detects lung alterations via small-angle X-ray scattering [1]
- Human-scale DFCT [2] image quality is limited due to **streak artifacts** from insufficient phase retrieval [3,4]
- Convolutional neural networks (CNNs) enable image enhancement using **multi-rotation ground-truth (GT) data** as target [5]

METHODS

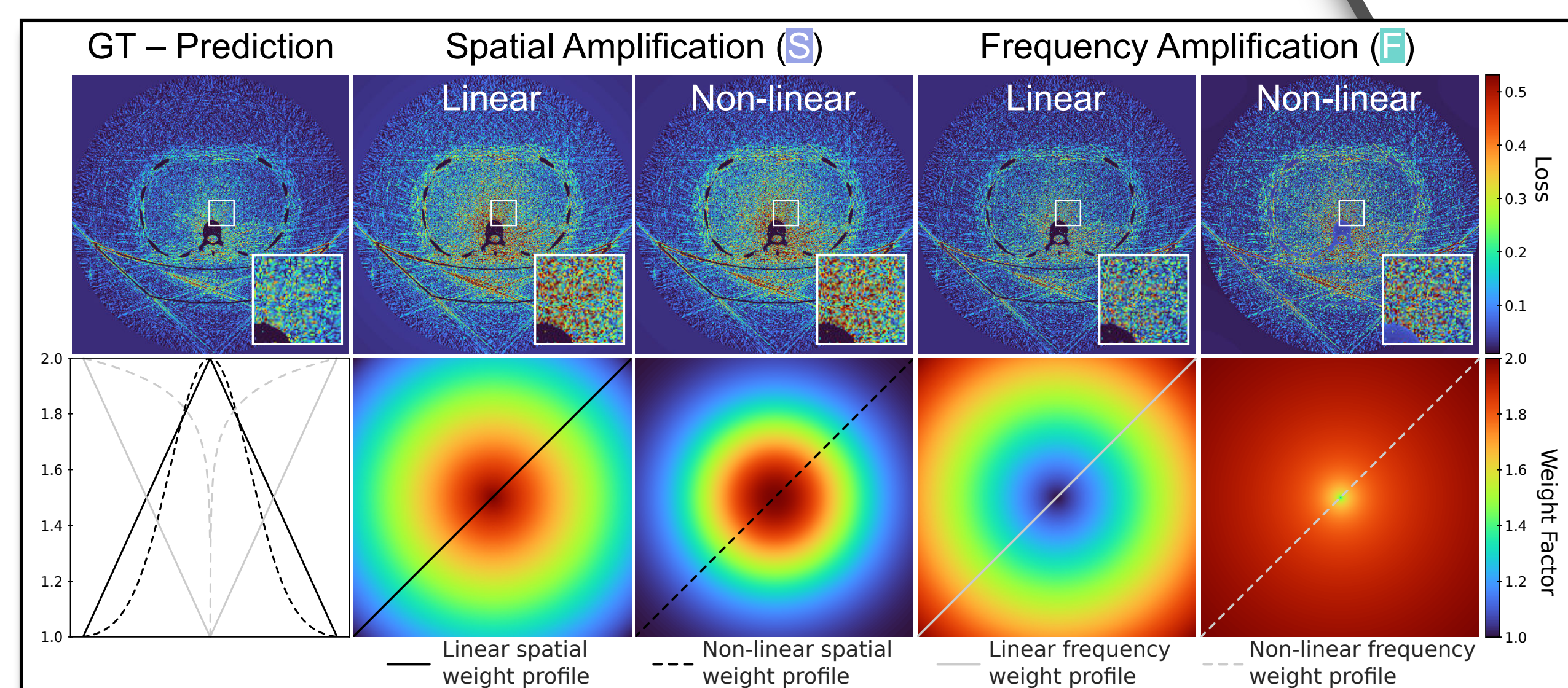
- Prototype gantry with rotating **Talbot-Lau interferometer**



- **2D U-Net** [6] trained on single-rotation scans (input) at clinically feasible dose and multi-rotation scans (GT) at up to 20-fold dose



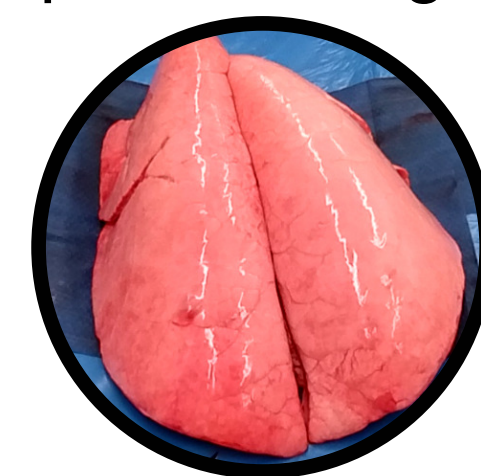
- **Structured loss amplification** boosts components of the 2D training loss map with designed spatial and frequency weights



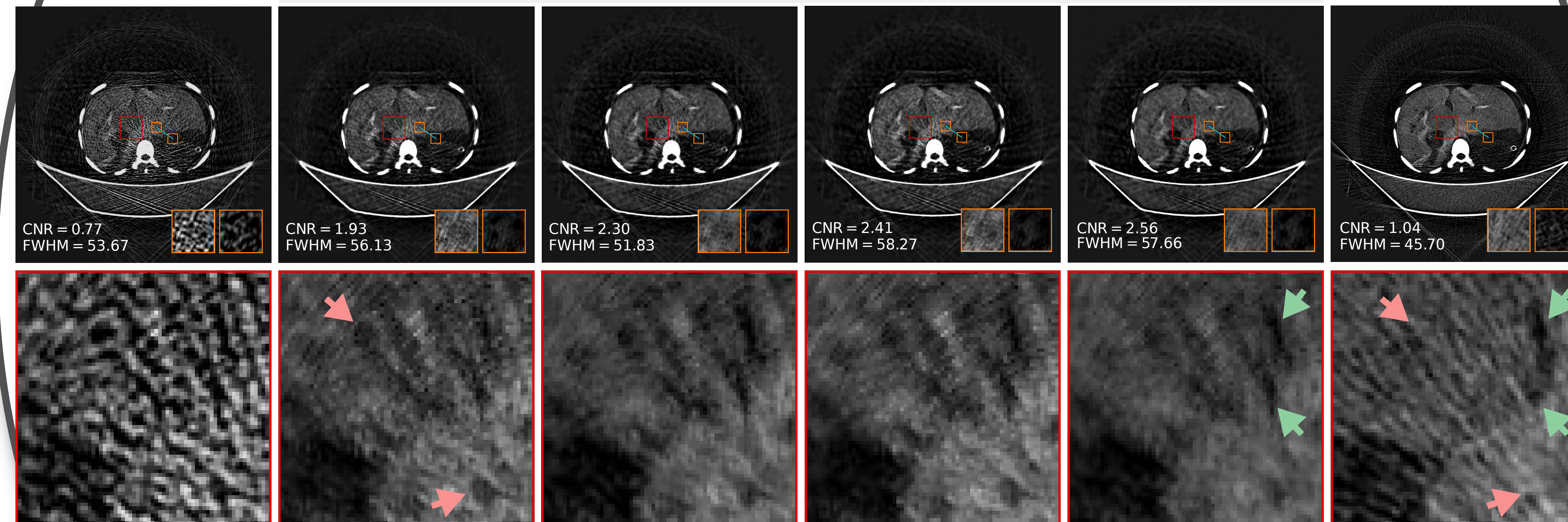
RESULTS

- **Spatial amplification (S)** with central weights increases both CNR and FWHM
- **Frequency amplification (F)** with high-frequency weights enhances CNR at slightly worse FWHM
- **Combined spatio-frequency amplification (S+F)** provides highest CNR at similar FWHM compared to baseline
- **CNR up to 33% higher** (2.52 ± 0.37 , $n = 10$) compared to baseline (1.89 ± 0.58) using L_2 loss

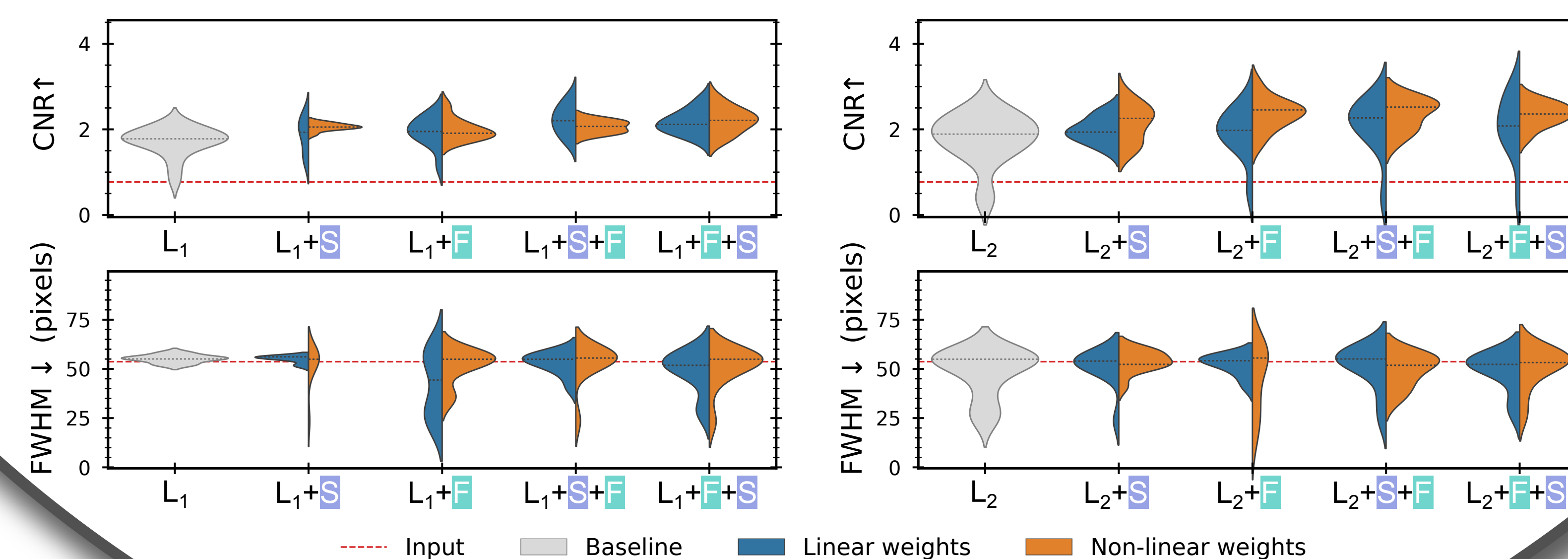
Ex vivo porcine lungs



Single-rot. (Input) U-Net (L_2) U-Net (L_2+S) U-Net (L_2+F) U-Net (L_2+S+F) Multi-rot. (GT)



CNR: Contrast-to-noise ratio FWHM: Full-width-at-half-maximum



Distribution of model performances across $n = 10$ repeated trainings

CONCLUSION

- Structured loss amplification **improves performance and robustness** of U-Net-based DFCT streak artifact reduction
- Formulation allows easy integration of **prior knowledge** into training objective
- U-Net suppresses both **streak artifacts and noise**

OUTLOOK

- Weighting schemes taking non-local, directed **signature of streaks** into account
- **GT enhancement** strategies to remove residual streaks in multi-rotation scans
- **Reader study** to validate improved DFCT quality for pulmonary imaging

REFERENCES

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